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# =====
# AI PRACTICAL 3
# Sorting Algorithms using Knowledge Representation
# =====

# STEP 1 : CREATE LIST

arr = [64, 34, 25, 12, 22, 11, 90]

print("Original Array:")

print(arr)


# =====
# BUBBLE SORT
# =====

n = len(arr)

for i in range(n):
    for j in range(0, n-i-1):

        # REASONING CONDITION

        if arr[j] > arr[j+1]:

            # SWAP ELEMENTS

            temp = arr[j]

            arr[j] = arr[j+1]

            arr[j+1] = temp


# STEP 2 : DISPLAY SORTED ARRAY

print("\nSorted Array:")

print(arr)

```